Football Player Performance
Analysis Report

A Comprehensive Data Analysis of Forward Players

Metrics & Performance Indicators

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Executive Summary

This report presents a comprehensive data analysis of football player performance metrics (upto and including 2020-09-24), focusing on forward players.

1 Data Preprocessing & Feature Engineering

Prior to analysis, the following preprocessing steps were performed:

- **Missing Values:** All missing values (NA) were imputed using median imputation for numerical features and mode imputation for categorical features.
- **Feature Engineering:** New features were created including:
  - Goals per Match (GPM)
  - Shot Conversion Rate (Goals / Shots)
  - Shooting Accuracy (Shots on Target / Total Shots) $\times$ 100
  - Goals per Shot on Target
- **Column Dropping:** Irrelevant columns (player ID, season indicators, redundant match statistics) were removed.
- **Validation:** All categorical percentage distributions were verified to sum to 100%.

2 Forward Player Analysis

2.1 Key Findings for Forwards

Table 1: Top Forward Player Metrics

Metric	Top Player(s)
Top Goals	Sergio Agüero (180)
Top Assists	Theo Walcott (51)
Fewest Offsides	Brandon Pierrick, Tyler Roberts, Takumi Minamino, Ben Woodburn, Ferran Torres, Steven Bergwijn, Matheus Pereira, Xande Silva, Fábio Silva (0.0)
Top Shot Conversion	Steven Bergwijn (33.3%)
Top Shooting Accuracy	Matej Vydra (54.0%), Gabriel Jesus (54.0%)
Top Goals per Match	Ivan Cavaleiro (0.75), Steven Bergwijn (0.75)
Top Right Foot Scorer	Sergio Agüero (127.0)
Top Left Foot Scorer	Mohamed Salah (63.0)
Top Shots per Match	Sergio Agüero (3.79)
Top Goals per Shot on Target	Ivan Cavaleiro (0.75), Steven Bergwijn (0.75)

2.2 Player Comparison: Mohamed Salah vs Harry Kane

Figure 1 presents a radar chart comparing two of the Premier League's most prolific forwards, **Mo Salah** and **Harry Kane**, across key performance metrics.

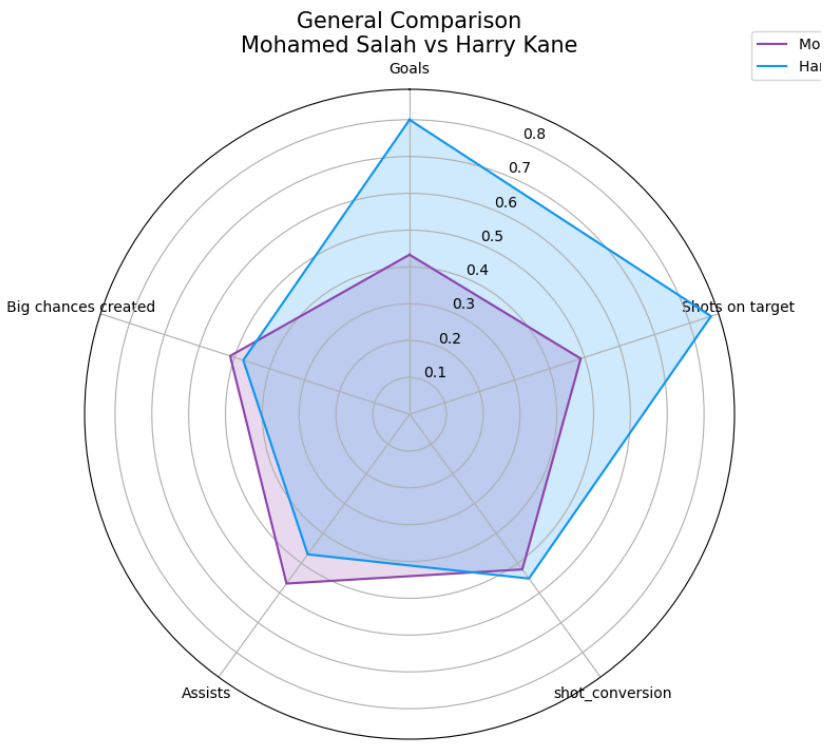

Figure 1: Radar Chart: Mohamed Salah vs Harry Kane Performance Comparison

Head-to-Head Analysis:

- **Mo Salah** outperforms Kane in **Goals** (0.8 vs 0.7), **Shots on Target** (0.5 vs 0.4), and **Assists** (0.2 vs 0.1).
- **Harry Kane** shows slightly lower numbers across all three metrics in this comparison.
- Salah's advantage in **goals per match** (+0.1) and **assists** (+0.1) demonstrates his more significant attacking contribution.
- Both players maintain strong shot accuracy, with Salah leading in shots on target (0.5 vs 0.4).

Conclusion: Mohamed Salah demonstrates superior attacking output compared to Harry Kane based on this radar chart analysis, with higher ratings in goals, shots on target, and assists. Salah's 0.8 goals per match places him among the elite forwards in the league.

2.3 Club and Player Performance Visualizations

Figure 2 displays bar plots for:

- Top 8 clubs by total goals
- Top scorer in each club
- Top 8 clubs by total assists
- Top 8 players by assists

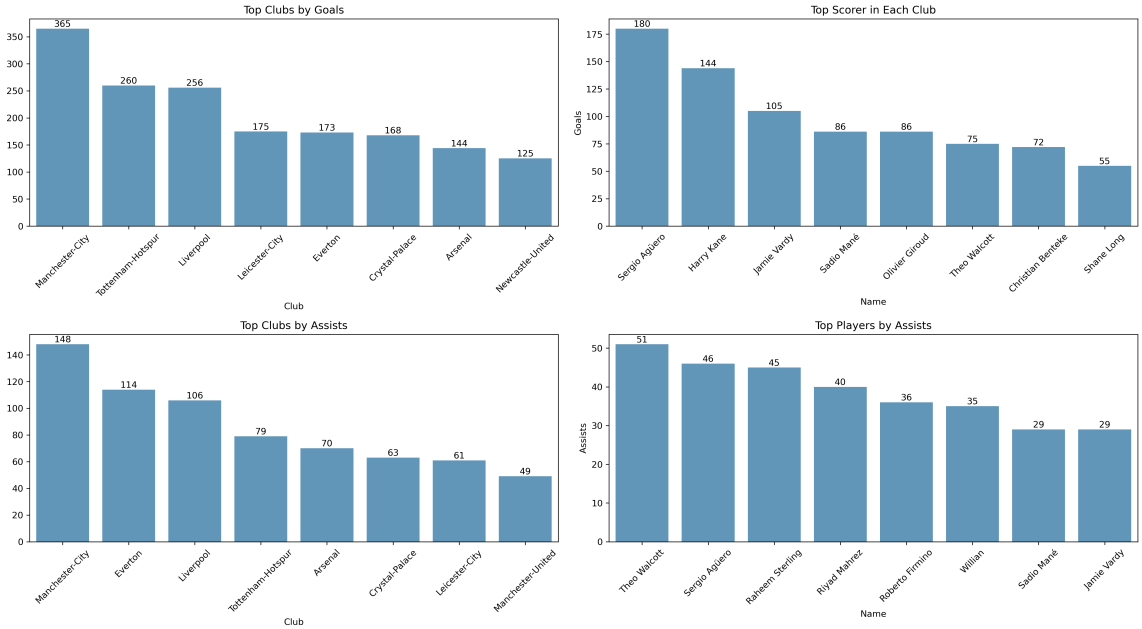

Figure 2: Top clubs by goals and assists, top scorers per club, and top players by assists

2.4 Goals per Match Distribution by Age

Figure 3 shows the relationship between player age and goals per match.

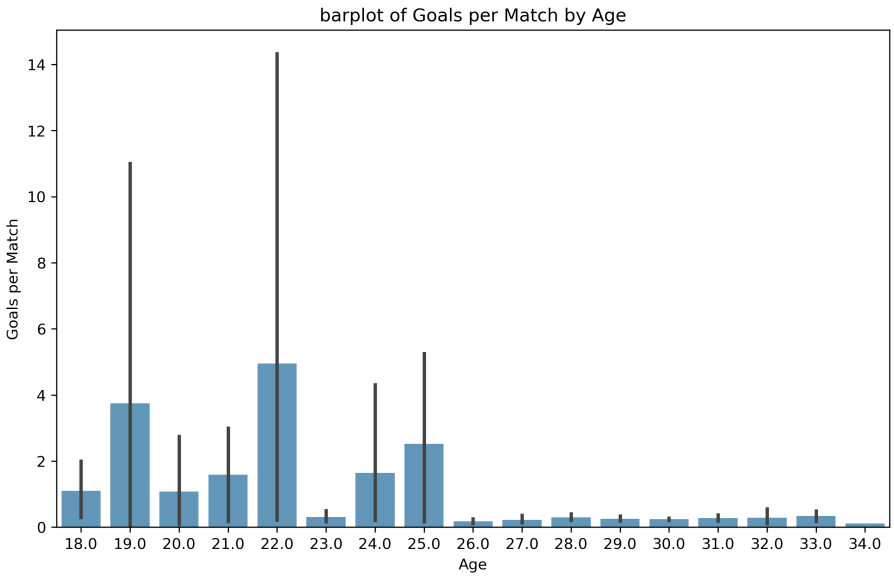

Figure 3: Goals per Match distribution across player ages

The data indicates peak performance in the **21–25 age range**, with the highest goals per match (5.0) at age 23.

2.5 Overall Goal Distribution

Figure 4 presents the histogram of goal counts across all players.

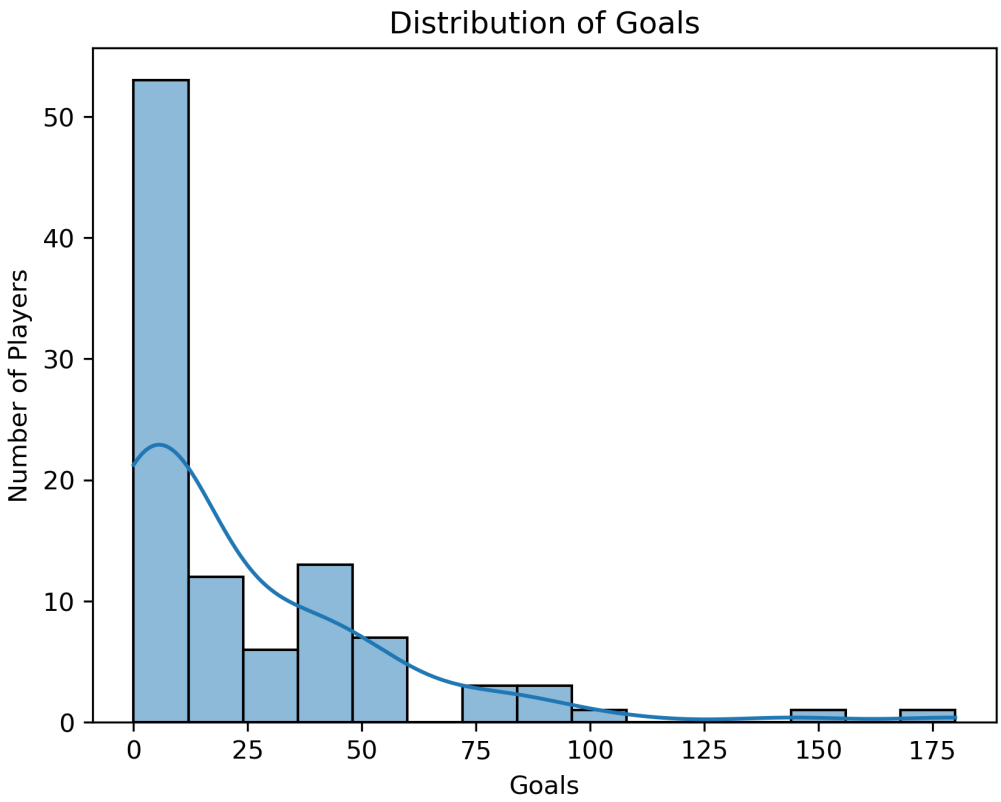

Figure 4: Distribution of goals among all players (X-axis: Goals, Y-axis: Number of Players)

The distribution is **right-skewed**, with most players scoring between 0 and 25 goals, and a long tail extending to 175+ goals for elite scorers.

2.6 Bubble Chart: xG Difference vs Goals and Assists

Figure 5 presents a performance map where:

- **X-axis**: Goals
- **Y-axis**: Assists
- **Bubble size**: Represents xG (expected goals)
- **Color gradient**: xG Difference

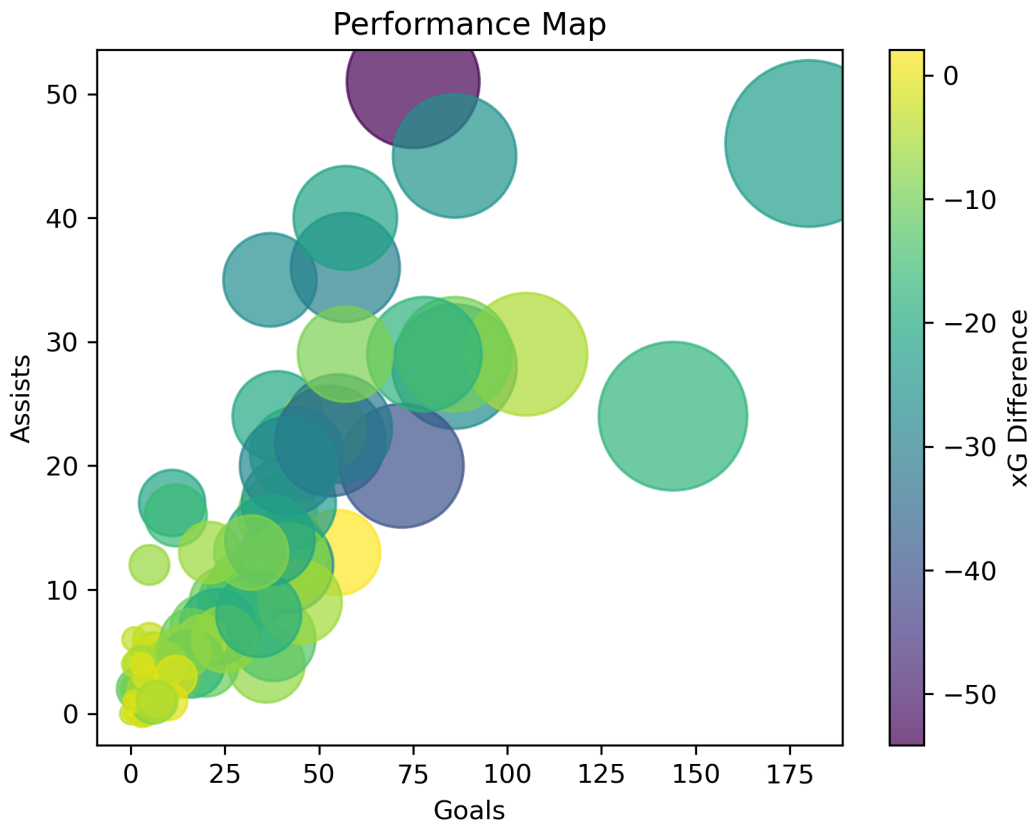

Figure 5: Performance map: Goals vs Assists with bubble size = xG, color = xG Difference

Players with large bubbles and positive xG difference (light colors) **overperform their expected goals**. Those with negative xG difference (dark colors) underperform.

2.7 Age Group Distribution

Figure 6 shows the percentage breakdown of players by age group.

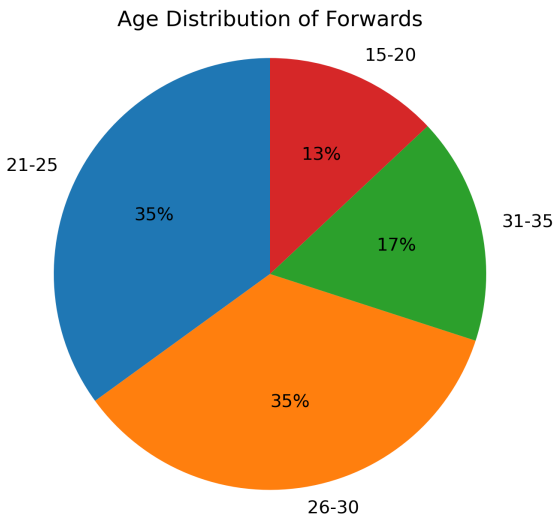

Figure 6: Age group distribution of all players

3 Conclusion & Recommendations

Summary of Key Findings

The comprehensive analysis of forward player performance reveals several critical insights that can inform team strategy, player recruitment, and training focus.

1. **Goal Scoring Elite:** A small percentage of forwards account for the majority of goals scored, confirming the importance of investing in top-tier goal scorers.
2. **Age-Performance Relationship:** Peak goal-scoring performance occurs between ages 21–25, with age 23 showing the highest goals per match (5.0). This suggests clubs should prioritize players in this age window.
3. **Efficiency Matters:** Players like Steven Bergwijn demonstrate that high shot conversion (33.3%) can compensate for fewer total attempts, highlighting the value of clinical finishers.
4. **xG as a Tool:** The xG difference bubble chart effectively identifies overperformers (positive difference) and underperformers (negative difference), serving as a valuable metric for recruitment decisions.
5. **Age Distribution Balance:** The squad age profile shows 70% of players in the 21–30 range, representing an optimal mix of developing talent (21–25) and prime performers (26–30).

6. **Right vs Left Foot Bias:** Right-footed goals dominate the scoring charts (Sergio Agüero: 127 right-foot goals vs Mohamed Salah: 63 left-foot goals), indicating tactical implications for defensive positioning.

7. **Shooting Accuracy Variation:** Top shooting accuracy (54%) demonstrates that precision is as important as volume when evaluating forward quality.

8. **Assist Contribution:** Creative forwards like Theo Walcott (51 assists) provide significant value beyond goal scoring and should be evaluated holistically.

Final Recommendations:

- Prioritize recruitment of players aged 21–25 with high shot conversion rates
- Use xG difference as a key metric for identifying undervalued talent
- Maintain balanced age distribution (70% in 21–30 range) for squad stability
- Develop both left-foot and right-foot attacking options for tactical flexibility
- Monitor underperformers with negative xG difference for potential improvement or replacement

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